

Post-Training-Associated Reorganization of Local Expert Competition in Sparse Mixture-of-Experts Models

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Abstract

Sparse mixture-of-experts (MoE) routers map continuous score profiles to discrete top- k expert sets. We ask whether post-training-associated routing change is mainly a redistribution of aggregate expert use or a reorganization of the local competitions that determine token-level selection. Across 512 GSM8K prompts, Qwen Base, A3B, and Thinking have similar median effective expert counts (50.0, 51.9, and 49.3) and nearly unchanged context specialization, yet their token-level top-8 sets and cutoff neighborhoods turn over substantially. Common-Trace Replay teacher-forces the same 64 Thinking-generated sequences through all three checkpoints, ruling out generated-text differences as the explanation: relative to Base, early-to-late top-8 Jaccard falls from .654 to .631 for A3B and from .623 to .581 for Thinking. We decompose this reorganization into continuous score profiles, shared-ID rank order, cutoff neighborhoods, and selected identities. Direct Thinking–A3B shared-ID score-rank agreement falls from .792 to .748 while top-8 and rank-6:10 Jaccard fall from .691 to .643 and .384 to .335, even as the scalar margin ratio remains near one. Under cross-fitted global-permutation controls, functional-fingerprint matching improves expert correspondence but fails to recover held-out local routing; matching optimized directly for token routing is nearly identity and also provides negligible recovery. Kanana, Nemotron, and Gemma4 reproduce the qualitative separability of score, rank, margin, and identity, but not one universal magnitude or positional direction. Finally, signed normalized margin erosion $\mathcal{R} = \epsilon|g|/\gamma$ serves as a functional assay: prospective high-risk interventions induce their targeted transition substantially more often than matched-low controls, and restoring changed expert IDs suppresses a median 99.7% of next-token KL among observed transition records. Routing organization and perturbation vulnerability are nevertheless distinct: a balanced pilot does not resolve an interaction between them. Thus post-training-associated changes can leave coarse utilization intact while reorganizing local expert competition; their observable decomposition transfers across architectures, whereas direction and scale remain checkpoint- and context-dependent.

1 Introduction

Sparse MoE routers create a discrete expert-selection mechanism: a continuous score profile is converted into a top- k identity set at every token (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Post-training can change model behavior substantially, but it is unclear whether it mainly redistributes aggregate expert usage or reorganizes the local competitions that determine individual routing decisions. Aggregate load, entropy, and frequency cannot answer this question: a checkpoint can preserve coarse utilization while changing which experts participate and which experts lie near a routing cutoff.

We ask: at what organizational resolution does routing change accompany post-training in sparse MoE models? Our answer has three parts. First, Qwen Base, A3B, and Think-

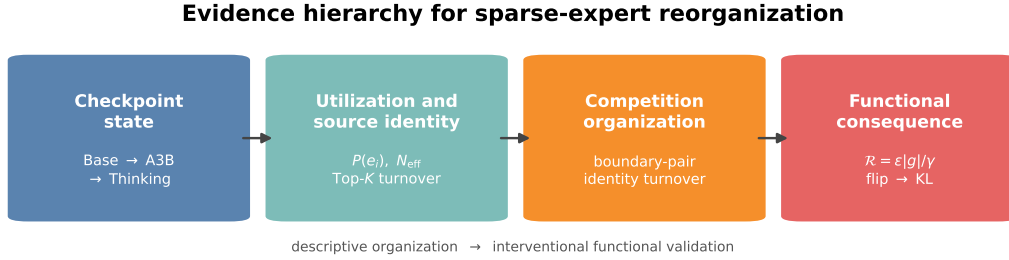


Figure 1: Evidence hierarchy. We compare checkpoint-associated utilization, source identity, and boundary competition before using local interventions to test whether the observed reorganization has a functional consequence. Natural trajectory dynamics and same-content replay then separate temporal structure from checkpoint-associated routing reorganization.

ing preserve coarse utilization and context specialization while changing token-level expert identity and cutoff-neighborhood membership. Second, Common-Trace Replay fixes the exact Thinking-generated sequence across the three checkpoints and shows that this reorganization is not explained by different generated text. Third, we empirically decompose routing change into continuous score profiles, shared-ID rank order, scalar cutoff margin, cutoff-neighborhood membership, and selected identity. This decomposition is measurable in Qwen, Kanana, Nemotron, and Gemma4, while its magnitude and positional direction are architecture-, checkpoint-, and context-dependent. We do not claim that score change causally produces the observed identity change; the contribution is to resolve these observables and show that they decouple empirically.

Figure 1 summarizes this evidence hierarchy. Our primary case study is the observational Qwen Base → A3B → Thinking progression; Kanana, Nemotron, and Gemma4 test which parts of the decomposition transfer across architectures. We then separate organization from functional vulnerability. Scaling a realized post-gating contribution by $1 + \delta_i$ and propagating the perturbed hidden state through later routers defines the normalized margin-erosion assay $\mathcal{R} = \epsilon|g|/\gamma$. This intervention-based risk is a functional probe of the resulting routing state, not a proxy definition for reorganization.

Our contributions are:

- Routing reorganization beyond tested global summaries. Similar coarse utilization can conceal substantial token-level changes. A stationary relation graph is unstable, functional-fingerprint matching does not recover routes, and the token-routing-optimal permutation is nearly identity. The tested aggregate utilization, stationary graph, and global relabeling do not recover the local organization.
- Score-to-identity decomposition. Common-Trace score replay decomposes local reorganization into score-profile, rank-order, margin, cutoff-neighborhood, and top- k identity measurements. Their empirical separability transfers across Qwen, Kanana, Nemotron, and Gemma4, while their magnitude and positional profile do not obey one universal law.
- Functional separation. Organization and perturbation vulnerability are distinct properties. The intervention-based $\mathcal{R}$ assay measures the latter, prospectively predicts targeted routing transitions, and connects observed transitions to output divergence without serving as a proxy for identity reorganization.

2 Related work

Sparse mixture-of-experts models. Sparsely gated layers increase capacity without proportional per-token computation (Shazeer et al., 2017). GShard and Switch Transformers developed scalable routing systems (Lepikhin et al., 2021; Fedus et al., 2022), and later work studied stable transfer at scale (Zoph et al., 2022). This line primarily motivates scalable

selection and load control; aggregate routing frequency does not determine whether the same experts compete at aligned token sites after checkpoint adaptation.

Local geometry and interventions. Fisher information describes local second-order distributional change under parameter perturbations (Amari, 1998). Our coordinates are not model parameters; they are temporary dimensionless scalars on post-gating expert contributions. The discrete top- k router creates piecewise-smooth regions separated by routing boundaries, whose effect we measure explicitly.

Expert interpretation and routing boundaries. Prior work has examined expert specialization, routing frequency, and modular structure. We focus on a complementary organizational question: after post-training, are the experts that compete at individual routing boundaries still organized in the same way? We distinguish aggregate utilization from token-level identity and cutoff-neighborhood structure, then use intervention only to test the separate functional vulnerability of the resulting local state. Unlike work that assigns a global interpretation to an expert ID, we treat organization as context- and site-conditioned and explicitly test whether a stationary graph or one global expert permutation is sufficient. Unlike work on router load shifts, our principal comparison holds generated content fixed and resolves score, rank, cutoff-neighborhood, and selected-set observables. The contribution is therefore the resolution at which routing change is measured, not merely the observation that checkpoint routing frequencies differ.

3 Measuring local expert competition

For router scores $r_{\ell,t,i}$, let $A_{\ell,t}$ be the selected top- k expert set and let $B_{\ell,t}$ be a five-expert neighborhood centered on the selection cutoff (ranks 6–10 in our reported experiments). The local safety margin is

$$\gamma_{\ell,t} = r_{\ell,t,(k)} - r_{\ell,t,(k+1)}. \quad (1)$$

For aligned sites from checkpoints or rendered contexts a and b , we report top- k Jaccard $J(A^a, A^b)$, cutoff-neighborhood Jaccard $J(B^a, B^b)$, the Spearman correlation of score ranks for expert IDs shared by the retained top lists, and the geometric margin ratio γ^b/γ^a . These are distinct observables: the scalar margin measures only the rank- k /rank- $(k+1)$ gap, while rank and set metrics measure broader score ordering and identity. Cross-family score values are never treated as calibrated on a common numerical scale. We use the tuple

$$\mathcal{C}(x, \ell, t) = (A(x, \ell, t), B(x, \ell, t), r_B(x, \ell, t), \gamma(x, \ell, t)) \quad (2)$$

as shorthand for local competition organization. The paper compares $\mathcal{C}^{\text{Base}}(x, \ell, t)$ with post-trained states at aligned contexts and sites; it does not assume that one global expert map $E_i^{\text{Base}} \mapsto E_j^{\text{Post}}$ is sufficient. Here r_B denotes scores restricted to the cutoff neighborhood; full retained score profiles are used only as supporting representations.

3.1 Functional vulnerability assay

Contribution-space intervention. For one MoE layer and token, let $h \in \mathbb{R}^d$ be the layer input, $f_i(h)$ expert i 's output, and $g_i(h)$ its normalized top- k gate. The realized contribution and baseline output are

$$u_i = g_i(h)f_i(h), \quad z_0 = b + \sum_{i=1}^E u_i. \quad (3)$$

We introduce $\delta \in \mathbb{R}^E$ and compute

$$z_\delta = b + \sum_{i=1}^E (1 + \delta_i)u_i, \quad \Delta z = U\delta. \quad (4)$$

The intervention is after gate normalization. It changes no expert input or parameter, performs no second normalization, and preserves the target layer's baseline expert identities. Inactive experts have zero local score. Downstream transformer and router layers execute naturally on the perturbed hidden state.

Supporting response geometry. We also evaluated a Fisher-like response matrix over contribution-space coordinates. It motivates the intervention but does not provide a robust global summary; its definition, validation, and negative robustness results are moved to Appendix A.

Routing-transition diagnostic. For downstream layer ℓ and token t , let r denote the scalar score that the implementation actually ranks to select experts, and let $\gamma = r_{(k)} - r_{(k+1)}$ be the baseline top- k boundary gap. For modified scores r' , define

$$\rho_{\ell,t} = \frac{\|r'_{\ell,t} - r_{\ell,t}\|_{\infty}}{\gamma_{\ell,t}}, \quad \rho(x, \delta) = \max_{\ell,t} \rho_{\ell,t}. \quad (5)$$

A flip occurs when any monitored downstream top- k set changes. Because $\rho(x, \delta)$ is a maximum over a forward pass, it is a record-level risk summary, not an event-level calibrated probability or causal threshold.

Baseline-only signed router-margin risk. The observed displacement in Eq. equation 5 is useful diagnostically but is not available before intervention. We instead differentiate the baseline boundary margin. Thus the same r is used for both ρ and prospective risk; raw logits are not assumed to determine selection. Qwen uses its native top-8 ranking score; Nemotron and Kanana use $r_i = \sigma(q_i) + b_i$ with a rank-6/rank-7 gap; Gemma uses the FP32 softmax probability passed to top-8, before selected- weight renormalization. Contribution weights are treated separately from ranking scores. For source expert i and downstream event (ℓ, t) , let

$$g_{i \rightarrow \ell,t}(x) = \left. \frac{\partial \gamma_{\ell,t}(x; \delta_i)}{\partial \delta_i} \right|_{\delta_i=0}, \quad \mathcal{R}_{i \rightarrow \ell,t}(x; \delta_i) = -\delta_i \frac{g_{i \rightarrow \ell,t}(x)}{\gamma_{\ell,t}(x)}. \quad (6)$$

The sign of δ_i is chosen to erode the fixed baseline rank- k versus rank- $(k+1)$ margin. With magnitude ϵ , the resulting nonnegative score is $\mathcal{R} = \epsilon|g|/\gamma$. We also report the intrinsic normalized sensitivity $|g|/\gamma$, which removes the chosen intervention scale. A first order prediction crosses the boundary when $\mathcal{R} \geq 1$.

This ratio has a direct interpretation: $\epsilon|g|$ is the locally predicted perturbation-induced boundary displacement, while γ is the available routing safety buffer. A transition therefore requires two ingredients—source susceptibility and boundary vulnerability—whose dimensionless product is tested directly in the prospective component ablation. Equivalently,

$$\log \mathcal{R} = \log \epsilon + \log |g| - \log \gamma. \quad (7)$$

Checkpoint or context variation can therefore act through boundary-margin geometry, source-to-router derivative sensitivity, or both. The cross-family experiments test these channels separately.

This definition separates three objects. $D_{\ell,i} = \mathbb{E}[s_{\ell,i}^2]$ is a marginal source-output sensitivity; γ^{-1} is destination vulnerability; and $g_{i \rightarrow \ell,t}$ is the directed source-to-boundary relation. Only their signed, context-specific combination predicts whether this source perturbation consumes this boundary’s safety margin.

From static relations to local vulnerability. A natural but stronger hypothesis is to aggregate $-g/\gamma$ over prefixes into a global directed graph whose nodes are (ℓ, i) . We test this hypothesis rather than assume it. Its failure motivates no post-hoc graph repair: the intervention object is a local vulnerability score $\mathcal{R}_{i \rightarrow \ell,t}(x)$ evaluated anew for each context, while the main organizational object is the natural routing identity and score profile.

Prospective targeted intervention and routing clamp. For a new prefix, we compute $\mathcal{R}$ from the baseline forward only. The high condition maximizes it over eight active source experts and all downstream events. Holding its destination (ℓ^*, t^*) fixed, a matched-low source has at most half the high risk when possible and is matched on marginal sensitivity, activation frequency, global and local gate weight, and expert-output norm. A third active expert is selected by a deterministic random seed. The exact top- k change at (ℓ^*, t^*) is primary; any

flip and next-token KL are secondary. Selection manifests are written and hashed before modified forwards.

Separately, for records already observed to flip, we restore baseline expert IDs at either the first changed event or all changed events while keeping every nonchanged event natural. This retrospective routing clamp asks how much observed divergence is removed by restoring expert choice. Conditioning on a flip prevents a population causal-effect interpretation.

4 Experimental protocol

Models and execution. The primary checkpoint is Qwen3-30B-A3B-Base. Cross-checkpoint experiments add Qwen3-30B-A3B and Qwen3-30B-A3B-Thinking-2507. These Qwen checkpoints have 48 MoE layers, hidden size 2,048, 128 experts, top-8 routing, and expert intermediate size 768. The Qwen source is zero-indexed layer 23 and destinations are layers 24–47. Cross-architecture experiments add NVIDIA Nemotron-3-Nano-30B-A3B-Base and its post-trained checkpoint. Nemotron has 52 hybrid Mamba/attention/MoE blocks, including 23 MoE blocks with 128 routed experts, top-6 routing, and one shared expert. Its source is physical block 27 (the 12th MoE block), and the 11 later MoE blocks are destinations; only routed contributions are perturbed. Gemma 4 experiments use 26B-A4B Base and IT weights. The executable model has 30 MoE text layers, 128 routed experts, top-8 routing, and no shared expert; its source is layer 14 and destinations are 15–29. We evaluate Base with the plain prompt and IT under the official thinking-OFF and thinking-ON chat renderings. Kanana-2 experiments use 30B-A3B Base and Thinking weights under the same plain prompt. Both have 48 decoder layers, with layer 0 dense and layers 1–47 MoE; each MoE layer contains 128 routed experts, top-6 selection, and one shared MLP whose width equals two routed experts. Qwen and Kanana use the same physical source layer 23 and destinations 24–47, matching normalized source depth while differing in attention, routing, and shared-expert structure. Expert parameters are layer-specific and IDs are not assumed semantically aligned across checkpoints or families. Models run in strict FP32, with TF32 disabled and FP64 KL evaluation. Seeds, configs, versions, scripts, selection hashes, and outcome hashes are stored.

Data and target protocol. We use GSM8K (Cobbe et al., 2021), pinned to revision 740312add88f781978c0658806c59bc2815b9866. A seeded permutation of the official train split produces 512 build and 128 disjoint held-out prefixes after filtering to 31–127 tokens. Answers are excluded from primary manifests. The prompt asks the model to solve the given problem and provide the answer. To avoid fixed-suffix routing collapse, the intervention is at the full-prompt token containing the question’s final non-punctuation content character; the evaluated distribution is the next token after the final prompt colon. The F build uses 512 prefixes; joint interventions use a fixed first 16-prefix held-out subset for the primary pilot. After that analysis was frozen, the same 40 $\epsilon = 0.01$ interventions were re-evaluated on all 128 held-out prefixes, producing 5,120 prefix records. This expanded run is a robustness and heterogeneity analysis, not an independent confirmation, and it does not access the sealed redesign confirmation split.

Two prospective G4-local splits contain 128 new prefixes each. The first is disjoint from all historical Stage 2 and reconstructed redesign IDs; the second also excludes the first G4-local split. Selection uses neither answers nor modified outputs. Cross-checkpoint and cross-architecture comparisons use the first prospective split, recomputing active experts, derivatives, destinations, and controls independently for every checkpoint/condition. Each uses its own 512-prefix marginal atlas. Gemma IT OFF and ON share weights but not input token sequences: OFF ends with an empty thought block, whereas ON adds a system turn containing `<|think|>`. We therefore treat them as two rendered contexts, not as an isolated causal intervention on thinking. For the cross-family score audit, we reuse the first 64 fixed prompt strings. Each checkpoint is tokenized independently and records its native top-list score profile; Gemma uses pre-top- k FP32 softmax probabilities and Nemotron uses $\sigma(q) + b$. Scores are compared by normalized prompt position, not by forced token identity. No modified forward is run for these replay artifacts. For trajectory analysis, Qwen Thinking greedily generates up to 512 tokens on 64 prompts. We store its generated token IDs, replay

the complete prompt plus continuation, and record top-8 identities at every layer. Common-Trace Replay then teacher-forces each identical stored token sequence through Qwen Base, A3B, and Thinking, recording top-8 IDs, normalized weights, the rank-9 outside expert, and the rank-6:10 neighborhood around the cutoff. The score replay additionally retains the top-16 router IDs and their sorted scores at every layer and token; all 64 token-sequence hashes match across checkpoints. Base and A3B do not naturally generate these traces; their measurements are conditional routing responses to Thinking-generated content.

Response construction and interventions. For each build prefix, we sample $y \sim p_0(\cdot | x)$ with a fixed seed, compute $s(x, y)$, and accumulate $ss^\top$. The primary estimate activates 85 experts at least once; 52 are activated at least 10 times. Joint interventions sample only from these 52 coordinates. We cross $|S| \in \{1, 2, 4, 8\}$ with $\epsilon \in \{0.01, 0.03, 0.05\}$, drawing 10 interventions per cell (120 total). The primary local analysis is $\epsilon = 0.01$: 40 interventions and 640 intervention–prefix records. The full grid gives 1,920 records for routing analysis; the frozen-intervention expanded analysis gives 5,120 records across 128 prefixes.

Metrics and uncertainty. At intervention level we report R^2 , Pearson r , Spearman ρ_s , MSE, and MAE. We use 20,000 paired bootstrap resamples of the 40 local interventions for full-minus-diagonal comparisons. The expanded analysis reports intervention, prefix-cluster, and two-way bootstrap intervals, each with 20,000 replicates. Routing groups use one-sided Mann–Whitney tests. Mean log-KL contrast and overall flip probability use 20,000 prefix-cluster bootstrap replicates. Wilson intervals on binned routing curves are descriptive and do not account for clustering. Trajectory summaries use medians across the 48 layers. All traces are divided into length-normalized early, middle, and late thirds of the observed generated continuation. The answer segment is reported only for traces containing the explicit closing thought delimiter. Temporal baselines compare top-8 Jaccard at lags 1, 2, 4, and 8 with random position pairs drawn within the same layer and trajectory; $8/(256 - 8) = .0323$ is shown only as a uniform-set approximation. Common-Trace agreement uses top-8 Jaccard, top-1 match, and rank-6:10 boundary-neighborhood Jaccard. Exact rank-8/rank-9 agreement is retained as a supplementary high-cardinality diagnostic. For the P2 mechanism pilot, one early-, middle-, or late-third midpoint is assigned to each trace by a deterministic cycle. At each of its 24 downstream layers we define boundary reorganization $O^B = 1 - J(B_{\ell,t}^{\text{Base}}, B_{\ell,t}^{\text{Thinking}})$ for the rank-6:10 neighborhoods, compute Base risk for the most vulnerable active source at layer 23, and form tertiles of O^B and $\log_{10} \mathcal{R}$. We freeze six sites per cell of the resulting 3×3 grid before running any modified forward. This is a balanced selected-site mechanism pilot, not a population estimate or randomized manipulation of O^B .

Protocol lineage and statistical units. Retrospective G2 uses 15,360 old held-out intervention–prefix records and is not independent confirmation. G3 uses 512 build prefixes and no modified forwards. Each prospective G4-local study uses 128 paired prefixes and three interventions per prefix; paired differences, 20,000-prefix bootstrap intervals, and exact McNemar tests are computed on prefixes. Cross-checkpoint effect differences use the same paired prefixes and are exploratory. Nemotron follows the same selection and outcome logic with a frozen family-specific numerical amendment: central finite differences with step 10^{-3} replace full-model forward AD, and margins are measured in the corrected routing-score space $\sigma(q) + b$. Gemma likewise uses central finite differences with step 10^{-3} and defines its margin in the exact pre-top- k FP32 softmax-probability space. Kanana also uses step- 10^{-3} central differences and the corrected $\sigma(q) + b$ rank-6/rank-7 score, while contribution weights use uncorrected sigmoid scores followed by top-6 normalization and scaling. Its shared MLP is never intervened upon. Multi-step FD audits pass for Gemma and Kanana, whereas Nemotron’s small-step full vectors are cancellation-limited and fail the same stability gate; we therefore treat Nemotron’s within-model risk enrichment as valid at the frozen oracle but do not infer a derivative shift mechanism. Routing clamps use all 225 existing flip-positive records and remain retrospective. Two additional prospective P0 studies use a fresh disjoint 128-prefix split: 64 prefixes for a five-condition component ablation and 64 for 512 non-max candidate interventions. Both selection manifests are frozen and hashed before modified forwards. No negative gate is overwritten, and the redesign confirmation file re-

mains sealed. For the new utilization summary, we normalize each checkpoint’s source-layer activation frequency to p_i , report $H = -\sum_i p_i \log p_i$, $N_{\text{eff}} = \exp(H)$, the standard Gini coefficient, and top-16 set Jaccard. These are layer-23 marginals over the 512-prefix build set; they do not estimate natural all-token co-activation or sequential transition matrices. All checkpoint and mode contrasts are observational associations. Expert IDs are compared only within a family and are not assumed semantically aligned across families. Qwen Common-Trace fixes token identity; other family score replays align normalized prompt position only. The permutation control learns a separate one-to-one expert assignment for each Qwen layer from one prefix half and evaluates it on the other; its fingerprints are routing-response proxies rather than direct expert-output evaluations.

5 Results

5.1 A natural-routing atlas localizes checkpoint reorganization

We record the selected expert IDs and normalized top-8 weights at every question token, every layer, and all 512 prefixes for Qwen Base, A3B, and Thinking: 35,368 prompt tokens and 13.58 million route assignments per checkpoint. Figure `effig:naturalatlas` shows three levels of organization. First, layer-wise effective expert count is nearly unchanged: its median is 50.0, 51.9, and 49.3. Second, unsupervised eight-cluster context specialization is also stable, with median normalized $I(E; C)$ of .01270, .01258, and .01255. Thus the atlas does not support the hypothesis that Thinking globally uses fewer experts or becomes more specialized on these prompt tokens.

Reorganization instead appears in identity and competition. Versus Base, the median layer-wise top-16 usage Jaccard is .778 for both A3B and Thinking, and top-1 expert identity is preserved on 64.7% and 69.4% of question tokens. The full top-8 set is preserved on only 13.5% and 17.6%, while the exact rank-8/rank-9 boundary pair is preserved on only 2.6% and 3.6%. Population edge sets are more stable but show the same ordering: median top-1000 Jaccard is .798/.794 for co-activation, .769/.779 for adjacent-token transitions, and only .532/.545 for boundary-pair competition. Post-trained states therefore preserve coarse load and much of the co-activation/transition skeleton while substantially reorganizing token-level membership and especially the marginal competition. A3B is not uniformly closer to Base than Thinking, so the atlas does not support a simple monotone training progression.

Generated-trajectory context. We greedily generate 64 Qwen Thinking continuations and use their stored token sequences for the Common-Trace experiment below. Descriptively, routing remains dynamic: median effective expert count increases from 60.8 to 68.2 from early to late observed positions, and lag-8 top-8 Jaccard (.156) remains above a matched within-trajectory shuffle (.132). Because most traces hit the 512-token cap and only Thinking is generated naturally, the full trajectory analysis is supporting rather than a checkpoint-level claim (Appendix).

5.2 Same-content replay isolates checkpoint-associated organization

Natural generation confounds checkpoint identity with generated text. We remove that content mismatch by teacher-forcing the same 64 Thinking-generated token sequences through Base, A3B, and Thinking. This is a routing-response comparison under shared content, not a claim that Base or A3B would naturally produce those sequences. Substantial checkpoint differences remain. Relative to Base, median-across-layer top-8 Jaccard for A3B is .654, .643, and .631 in early, middle, and late generation; Thinking is less aligned at .623, .603, and .581. Its decline from early to late is also larger (.042 versus .023). Top-1 match remains around .61–.65 and does not show a monotone stage effect (Table 5).

The cutoff neighborhood shows particularly low agreement. Rank-6:10 neighborhood Jaccard changes from .334 to .322 for A3B and from .270 to .252 for Thinking. Exact rank-8/rank-9 pair agreement is only 1–2%, but we treat that as supplementary because exact pairs have high cardinality. Length-normalized curves corroborate the stage summaries: Thinking remains less aligned than A3B and reaches .566 top-8 Jaccard in the final bin,

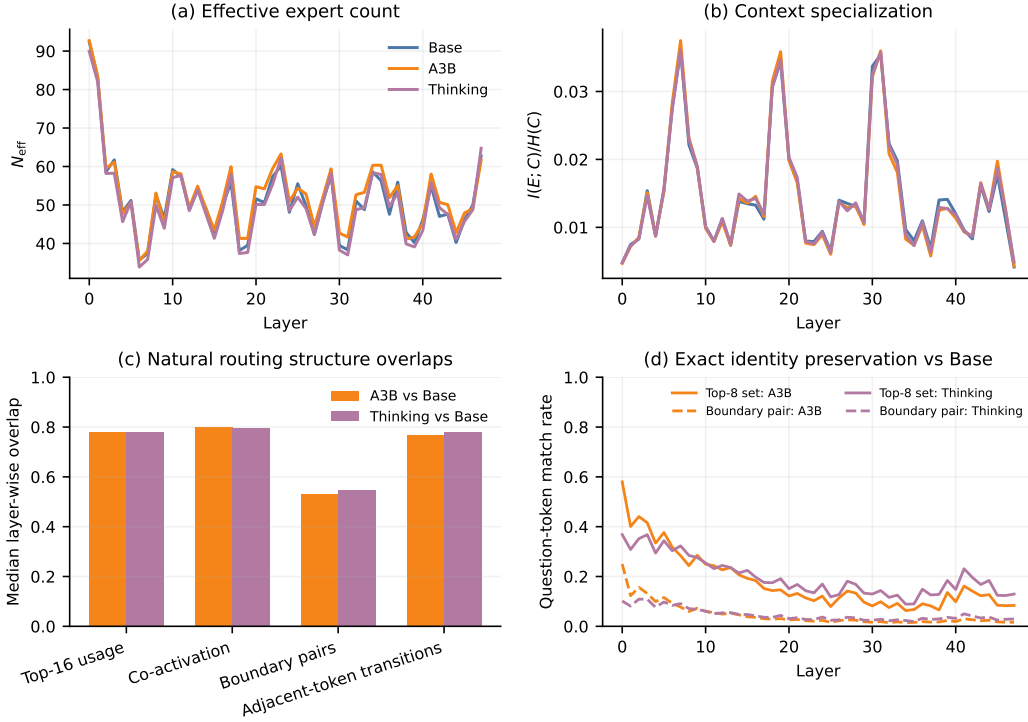


Figure 2: Natural-routing atlas over 512 GSM8K prompts. (a) Effective expert count and (b) checkpoint-independent context specialization are nearly unchanged. (c) Layer-wise population edge overlap remains highest for co-activation and adjacent-token transitions and lowest for boundary pairs. (d) Exact question-token expert identity rapidly diverges with depth, with the rank-8/rank-9 competing pair least preserved.

versus .616 for A3B. Therefore, holding prompt and continuation tokens fixed, checkpoint identity is associated with routing reorganization, and the Thinking state differs more from Base than ordinary A3B on these traces, especially at later observed positions.

5.3 Score-level decomposition of cutoff reorganization

Identity turnover alone does not say whether the router scores moved or merely reshuffled their local order. We therefore rerun the same-content replay while retaining the top-16 router scores (the existing ID-only replay is unchanged). For each checkpoint we report the top-8/top-9 margin γ , the centered and scale-normalized score-profile distance at ranks 6–10, and Spearman agreement of score values for expert IDs appearing in both top-16 lists. The latter is computed on shared expert IDs; Spearman on the already sorted rank slots would be tautologically one and is omitted.

Figure 3 gives the position-dependent decomposition. Thinking has geometric-mean cutoff-margin ratios of 1.040, 1.047, and 1.043 relative to Base in early, middle, and late generation, respectively. A3B has 0.995, 1.056, and 1.059, so its margin shift is less uniform. At the same time, Thinking’s shared-ID score-rank Spearman falls from .727 to .704 to .687, and its rank-6:10 Jaccard falls from .287 to .282 to .272. The corresponding A3B values are .753, .742, .727 and .334, .335, .322. Thus Thinking’s stronger late discrete turnover coexists with a modest margin expansion and progressive local rank reshuffling; the identity result is not explained by a single scalar margin shift. Raw score levels are interpreted only as same-architecture, same-content relative quantities because router-score parameterizations need not be globally calibrated across checkpoints.

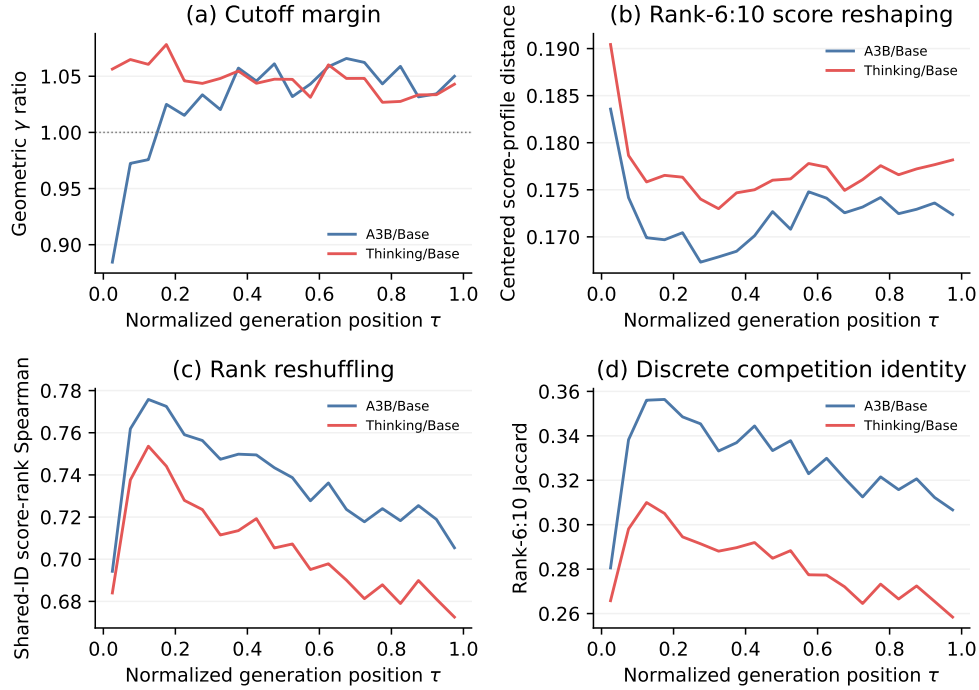


Figure 3: Score-level decomposition under Common-Trace Replay. (a) Thinking shows a modest, position-stable cutoff-margin expansion relative to Base, while A3B is less uniform. (b) Centered rank-6:10 score profiles are reshaped in both comparisons. (c) Spearman agreement of scores for expert IDs shared by the two top-16 lists declines with position, more strongly for Thinking. (d) The same positions exhibit discrete cutoff-neighborhood turnover. Rank-slot Spearman is excluded because both score vectors are sorted by construction.

Table 1: Score-level Common-Trace decomposition with Base. Values are medians across layers for the indicated generation segment. γ ratio is the geometric mean across site-level ratios; score-rank agreement uses shared expert IDs in the top-16 lists.

Checkpoint	Segment	γ ratio	Score-profile distance	Shared-ID ρ_s	Boundary Jaccard
A3B	Early	0.995	.172	.753	.334
	Middle	1.056	.170	.742	.335
	Late	1.059	.173	.727	.322
Thinking	Early	1.040	.179	.727	.287
	Middle	1.047	.175	.704	.282
	Late	1.043	.177	.687	.272

The third edge of the Qwen checkpoint triangle is available without another forward pass. Direct Thinking–A3B agreement also declines from early to late: top-8 Jaccard falls from .691 to .643, rank-6:10 Jaccard from .384 to .335, and shared-ID score-rank Spearman from .792 to .748. In contrast, the geometric cutoff-margin ratio changes only from 1.067 to .980. Thus the result is not only two separate distances from Base: A3B and Thinking themselves become less aligned over the observed continuation even though their scalar cutoff margins remain similar.

5.4 A global expert permutation does not explain turnover

Low same-ID overlap could arise if post-training merely permuted a stable set of expert roles. We test this alternative with two layer-wise Hungarian assignments, each learned on 32 prefixes and evaluated on the disjoint half. Functional- fingerprint matching improves

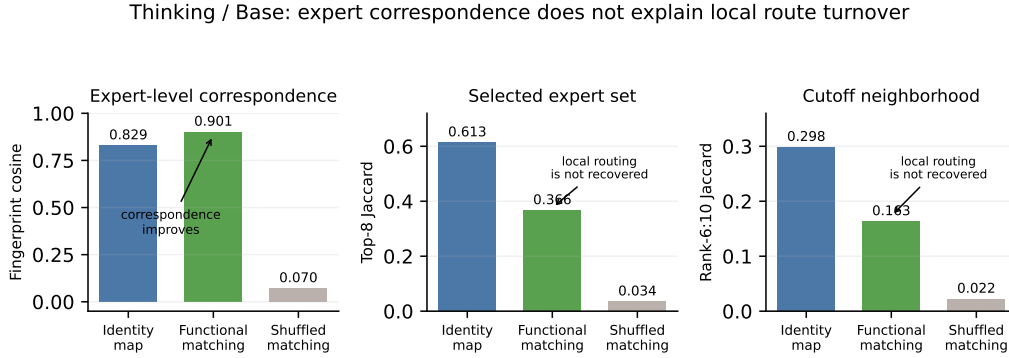


Figure 4: Three-level Qwen Thinking/Base permutation control. Hungarian assignments are learned from expert routing fingerprints on one prefix half and evaluated on the other. Functional matching raises held-out fingerprint cosine (.829 to .901), confirming expert-level correspondence, but lowers top-8 and cutoff-neighborhood overlap (.613 to .366 and .298 to .163). Shuffled matching shows the random-permutation expectation. Thus correspondence improves without recovering local routing. Separately, the routing-optimized map is nearly identity (88% fixed points) and changes held-out top-8 Jaccard by only .00011.

held-out routing-profile cosine for Thinking–Base from .829 to .901, yet top-8 and rank-6:10 Jaccard fall from .613 to .366 and .298 to .163. Thus expert-level routing correspondence is not token-level route correspondence. More directly, a second assignment optimizes binary top-8 activation-vector Jaccard on the training tokens. It is 88% fixed points, gains only about 5×10^{-4} over identity on its training objective, and changes held-out top-8 Jaccard by only .00011. The routing-optimal global permutation is therefore nearly identity and provides no substantive recovery. Figure 4 shows the primary Thinking–Base control; complete pairs, intervals, and objectives are in the appendix. These routing-derived fingerprints are not evidence of semantic or full input–output equivalence.

5.5 Cross-architecture score-level decomposition

Kanana. We next repeat only the four most diagnostic measurements in an independent MoE architecture. We teacher-force the same 64 plain GSM8K prompt strings through Kanana Base and Thinking, using their native tokenizer and actual choice score $r_i = \sigma(q_i) + b_i$. Kanana has top-6 routing, so γ is the rank-6/rank-7 gap; rank 6–10 remains the five-expert cutoff neighborhood. This replay contains 4,409 prompt tokens across 47 MoE layers and performs no intervention.

Table 3 shows that Kanana also has low cutoff-neighborhood identity agreement (.226–.245) despite higher top-6 set overlap (.602–.624), together with shared-ID score-rank agreement around .75. Its margin ratio is larger and nonmonotone (1.333, 1.463, and 1.300), and its identity/rank metrics do not exhibit Qwen’s late-position decline. The independent family therefore reproduces the continuous-score/discrete-identity decomposition but not a universal positional direction. We interpret this as architecture- and checkpoint-dependent heterogeneity, not a failed replication.

Nemotron and Gemma4. To test whether the decomposition was specific to the two preceding families, we repeat the same fixed-prompt normalized-position analysis for the existing Nemotron Base–Post checkpoints and for Gemma4 Base, IT-OFF, and IT-ON. Nemotron uses its corrected choice score $\sigma(q) + b$ and top-6 routing; Gemma uses the FP32 pre-top-8 softmax probability. Each comparison contains 64 identical prompt strings, but checkpoint-specific tokenizers, so these are normalized position comparisons rather than token-identity common traces.

The results in Table 2 and Figure 5 reinforce the heterogeneous interpretation. Nemotron Post/Base has top-6 Jaccard .384, .348, and .326 and rank-6:10 Jaccard .146, .132, and .123

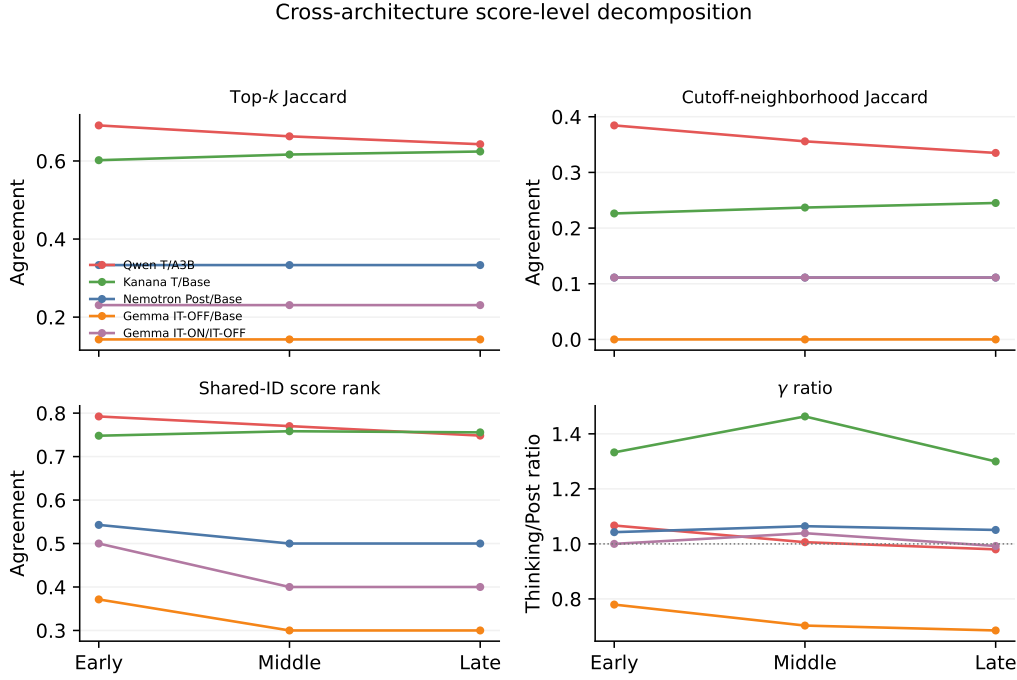


Figure 5: Cross-architecture score-level decomposition on fixed prompt strings. Qwen and Kanana use the generated/common or plain replay described above; Nemotron and Gemma use normalized-position matching with checkpoint-specific tokenizers. Across families, score-rank reshaping and cutoff identity are measurable separately, while margin magnitude and positional profiles are heterogeneous.

Table 2: Nemotron and Gemma4 score-level replication. Entries are medians over router layers for early/middle/late normalized positions. Top- k is 6 for Nemotron and 8 for Gemma; rank-6:10 is the corresponding five-ID cutoff window.

Pair	Segment	γ ratio	Shared-ID ρ_s	Cutoff Jaccard	Top- k Jaccard
Nemotron Post/Base	Early	1.043	.543	.146	.384
	Middle	1.064	.500	.132	.348
	Late	1.051	.500	.123	.326
Gemma IT-OFF/Base	Early	.779	.371	.056	.143
	Middle	.703	.300	.056	.143
	Late	.685	.300	.054	.143
Gemma IT-ON/IT-OFF	Early	1.000	.500	.111	.231
	Middle	1.039	.400	.111	.231
	Late	.992	.400	.111	.231

from early to late, while its margin ratio remains close to one (1.043, 1.064, 1.051). Gemma IT-OFF/Base has very low local identity agreement (top-8 Jaccard .143-.148 and cutoff-neighborhood Jaccard about .055) and a margin ratio that falls from .779 to .685. Holding Gemma IT weights fixed, IT-ON/IT-OFF has margin ratios 1.000, 1.039, and .992; its score-rank and cutoff identity profiles shift, but neither follows the Qwen late-decline pattern. Thus the robust cross-family statement is that continuous score profiles and discrete cutoff identities can be analyzed as separate quantities; their amplitude and positional direction depend on architecture, checkpoint, and rendered context.

The positional contrasts are not inferred from point estimates alone. A prefix-bootstrap over the 64 replay prefixes gives the early-minus-late intervals in Table 4. For Qwen, all three identity/rank contrasts exclude zero. Kanana has a small early-to-late increase in overlap for the two identity measures, while its score-rank agreement and margin ratio are nonmonotone; the data therefore do not support a common monotone positional trajectory.

Table 3: Direct Qwen Thinking–A3B comparison and minimal Kanana replication. Values are medians across layers. γ ratios are Thinking/A3B for Qwen and Thinking/Base for Kanana; top- k is 8 and 6, respectively.

Pair	Segment	γ ratio	Shared-ID ρ_s	Rank-6:10 Jaccard	Top- k Jaccard
Qwen Think./A3B	Early	1.067	.792	.384	.691
	Middle	1.006	.770	.356	.663
	Late	.980	.748	.335	.643
Kanana Think./Base	Early	1.333	.748	.226	.602
	Middle	1.463	.758	.237	.617
	Late	1.300	.756	.245	.624

Table 4: Prefix-bootstrap early-minus-late contrasts over 64 replay prefixes. Values are differences in the layer-median stage summaries; intervals contain 95% bootstrap replicates (20,000 draws). Positive values indicate larger early agreement.

Pair	Metric	Early – Late	95% CI
Qwen Think./A3B	Top- k Jaccard	.047	[.038,.056]
	Rank-6:10 Jaccard	.048	[.039,.057]
	Shared-ID ρ_s	.044	[.036,.051]
Kanana Think./Base	Top- k Jaccard	−.023	[−.034, −.014]
	Rank-6:10 Jaccard	−.016	[−.027, −.007]
	Shared-ID ρ_s	−.004	[−.012, .005]

Supporting global-summary tests. Earlier experiments tested whether a single full response matrix or stationary source-to-router graph could replace the local description. In a 16-prefix pilot, full and diagonal response predictors were similar ($R^2 = .837$ and $.841$); over 128 prefixes, pooled full-matrix R^2 fell to $-.088$. A preregistered static graph also failed edge-rank, top-edge, and out-strength stability gates. These negative results motivate local measurement but are not a second headline; full geometry, estimator, and graph diagnostics appear in the appendix.

5.6 Functional relevance and separation

Organization describes which competition is present; risk describes how easily that competition can be crossed. Retrospectively, signed maximum $\mathcal{R}$ reaches PR–AUC .974 on 15,360 records, but this motivates rather than replaces the prospective tests below.

Prospective validation on two unseen splits. G4-local tests the vulnerability of a selected local competition directly. For each new prefix, all selection features are baseline-only; the high, matched-low, and random conditions share the exact destination event and margin. The selection manifest is hashed before modified forwards. No matched-low selection is relaxed.

On the first independent split, high-risk sources flip the targeted event in 59/128 prefixes (0.4609), versus 13/128 (0.1016) for matched-low and 15/128 (0.1172) for random. The primary paired high-minus-low effect is 0.3594 with 95% prefix-bootstrap CI [0.2812, 0.4453]; all 46 discordant pairs are high-only (exact McNemar $p = 2.84 \times 10^{-14}$). On a second disjoint split, rates are 46/128 (0.3594), 12/128 (0.0938), and 16/128 (0.1250), with primary effect 0.2656 [0.1875, 0.3438] and 34 high-only discordant pairs ($p = 1.16 \times 10^{-10}$). Both independently frozen gates pass. The secondary fixed-effect summary is 0.3086 [0.2523, 0.3649], but $I^2 = 62.2\%$; we therefore emphasize study-specific estimates.

These are prospective targeted validations, not randomized treatment effects: source conditions are selected from baseline geometry. Nevertheless, matching fixes the destination and balances source marginals, selection precedes outcomes, and the exact preselected event is primary. The evidence therefore establishes predictive intervention utility of the relational

Table 5: Common-Trace Replay agreement with Qwen Base. Values are medians across 48 layers under identical Thinking-generated token sequences.

Checkpoint	Segment	Top-8 Jaccard	Rank-6:10 Jaccard	Top-1 match
A3B	Early	.654	.334	.640
	Middle	.643	.335	.650
	Late	.631	.322	.637
	Answer*	.618	.307	.642
Thinking	Early	.623	.270	.613
	Middle	.603	.263	.628
	Late	.581	.252	.608
	Answer*	.576	.246	.628

Table 6: Prospective exact targeted-event routing transitions.

Study	High	Matched low	Random	High-low (95% CI)
G4-local 1	0.4609	0.1016	0.1172	0.3594 [0.2812,0.4453]
G4-local 2	0.3594	0.0938	0.1250	0.2656 [0.1875,0.3438]

score as a vulnerability assay, rather than treating risk as a definition or proxy for identity reorganization.

Component necessity and non-max ordering. On a fresh disjoint split, sensitivity-only and margin-only selectors each produce 0/64 exact target transitions, while normalized $|g|/\gamma$ produces 32/64 versus 6/64 for a source-matched same-boundary control. On another 64 prefixes, all 12 transitions among 512 non-max candidates fall in the highest frozen-risk quintile. Thus, in this prospective pilot, neither component alone produces the targeted transition, whereas their normalized combination does; ordering is also not an artifact of selecting only the global maximum.

Routing restoration. We next analyze all 225 old E5-A records with an observed transition. Restoring all changed expert IDs suppresses a median 0.99677 of natural next-token KL; 96.9% of records suppress more than half and 88.4% suppress more than 90%. Restoring IDs and baseline weights yields a similar median (0.99661). Restoring only the first transition has the same median suppression; its KL equals the all-flip clamp exactly in 216/225 cases. Nine exceptions, including two negative first-clamp restoration fractions, show that later transitions can interact nonadditively. Zero-intervention clamps are exactly inert and natural flip counts are reproduced in 225/225 cases. Routing restoration therefore accounts for most divergence in observed transition records. Because the analysis conditions on flips and is retrospective, it is not a population causal-effect estimate.

Finally, a balanced 54-site $O^B \times \mathcal{R}$ pilot observes only two target transitions, both at high risk but at low and high reorganization. It does not resolve an interaction: boundary reorganization is not itself a vulnerability score. Full retrospective, component, non-max, clamp, and pilot tables are in the appendix.

Cross-family functional summary. We recompute active experts, derivatives, destinations, and controls independently on the same 128 prefixes for every condition. Local risk is prospectively enriched in all ten conditions, but between-condition shifts are heterogeneous. The high-minus-low effects are .359/.336/.219 for Qwen Base/A3B/Thinking, .172/.148 for Nemotron Base/Post, .125/.250/.203 for Gemma Base/IT-OFF/IT-ON, and .414/.289 for Kanana Base/Thinking. Only the Qwen Thinking-Base effect contrast is resolved; the other checkpoint/mode contrasts are not. These results support transfer of the vulnerability assay, not a universal robustness shift.

Selected-event geometry varies by family: Qwen Thinking is primarily margin-associated, Kanana Thinking is derivative-associated at stable margins, and Nemotron Post shows no resolved shift. Gemma changes both weights and rendered context across its conditions.

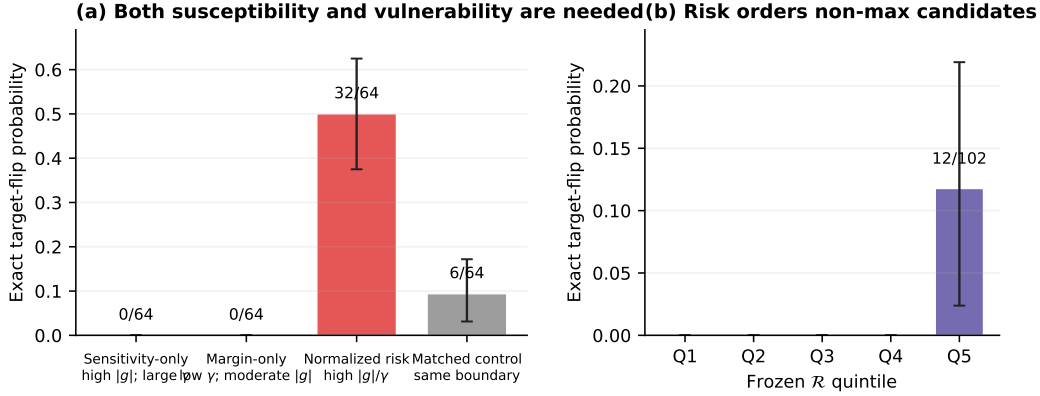


Figure 6: Prospective evidence for the normalized risk combination and against an artifact of maximum selection. (a) Sensitivity-only and margin-only selectors do not induce their targeted transition; $|g|/\gamma$ does and exceeds a same-boundary matched control. (b) For 512 candidates frozen without maximizing risk, all observed transitions fall in the highest risk quintile. Bars show prefix-cluster bootstrap 95% intervals.

Detailed geometry, calibration, common-site boundary turnover, and KL summaries are reported in the appendix.

6 Discussion

What changes? Coarse expert utilization can remain similar while local competition changes. Qwen Base, A3B, and Thinking have comparable effective counts and context specialization, yet aligned top-8 sets and cutoff neighborhoods turn over. The Common-Trace control makes this a checkpoint-associated organization result rather than a consequence of different generated text. Score replay further shows that margin, shared-ID rank, cutoff neighborhood, and selected identity are not interchangeable observables. The same decomposition is measurable in four families, but its direction and magnitude are not universal.

What is the right object? The evidence favors the context- and site-conditioned tuple $\mathcal{C}(x, \ell, t)$ over the tested global summaries. Aggregate utilization misses token-level turnover; a stationary source-to-router graph fails stability gates; functional-fingerprint matching improves expert correspondence without recovering routes; and the routing-optimal permutation is nearly identity with negligible gain. Expert ID continuity therefore contains signal, but a single global graph or relabeling is not sufficient to describe local organization. This is an empirical resolution claim, not an exhaustive impossibility result over all global summaries or a proof that measured score changes causally produce identity turnover.

What does it mean functionally? Organization and vulnerability answer different questions: $\mathcal{C}$ describes which competition is present, whereas $\mathcal{R}$ describes how easily a specified source perturbation crosses it. Prospective targeting, component ablation, and non-max ordering validate $\mathcal{R}$ as a transition assay; routing restoration connects observed expert replacement to output divergence. The small $O^B \times \mathcal{R}$ pilot does not establish an interaction, so reorganization is not itself a vulnerability score. Across families the assay transfers, while its operating distribution and checkpoint/mode contrasts remain heterogeneous. All training-state comparisons are observational; Gemma mode additionally changes rendered context.

7 Limitations

All primary interventions use GSM8K and source layer 23; layer and task transfer remain open. The natural atlas covers prompt-question token utilization, within-token co-activation, rank-8/rank-9 competition, and adjacent-token routing transitions, while the trajectory experiment covers generated tokens only for Qwen. Its context analysis uses unsupervised TF-IDF clusters rather than gold mathematical-skill labels. G4 establishes internal targeted routing transitions, not sequence correctness: next-token top-1 identity never changes in the cross-checkpoint intervention grid, and no interventions are applied along the generated traces. High and matched-low sources are baseline-selected rather than randomized, so results demonstrate prospective targeting utility rather than the causal effect of assigning a risk score.

Weight-state comparisons are exploratory. Qwen Base, A3B, and Thinking-2507, the two Nemotron states, Gemma Base/IT, and Kanana Base/Thinking lack controlled training trajectories. Their differences are checkpoint associations, not causal effects of reasoning-oriented post-training. Expert IDs are not semantically aligned across checkpoints or families. The Hungarian controls address only a single global layer-wise permutation under routing-derived fingerprints or training-token activation correspondence; they do not rule out context-dependent correspondences, mixtures of expert roles, or drift in the experts' full input-output functions. Moreover, each condition independently selects its highest-risk target event. Geometry ratios therefore compare two risk-selected operating points, not an unconditional shift over all events. Our common-site audit freezes (ℓ, t) , but the actual rank-boundary expert pair almost always changes across weight states; it therefore aligns a physical router site rather than the same decision boundary. The 40.5 and 52.0 state-specific site-margin ratios reported in the appendix must not be interpreted as expansion of a shared boundary. Existing basis artifacts do not serialize nonselected event derivatives, so we explicitly avoid the stronger global-distribution claim. Gemma OFF/ON uses the same weights but different chat-template tokens and different prediction contexts; it is not an isolated causal thinking variable. Cross-family numerical levels are not directly comparable because routing top- k , source depth, shared- expert structure, tokenizer, architecture, and derivative oracle differ. The Kanana, Nemotron, and Gemma score replications are deliberately minimal: they use fixed prompt strings rather than a shared generated continuation and use each checkpoint's tokenizer. Their normalized-position comparisons test transfer of the score-level decomposition, not cross-family equality of score calibration or token-level alignment. Gemma IT-OFF/ON additionally changes the rendered context at fixed weights.

G2 and routing clamps reuse observed transition data and remain retrospective. Routing restoration is conditioned on flipping and is not an additive natural direct/indirect effect or population causal-effect estimate. The local quadratic pilot is small, while its 128-prefix expansion fails pooled absolute-magnitude robustness. Full-matrix, static-graph, and estimator redesign failures are retained rather than repaired post hoc. Finally, all findings are conditional on observed routing support, GSM8K prompts, and the four evaluated model families. The non-max experiment additionally conditions candidate sampling on the 16 smallest-margin events per prefix; it supports prospective ordering in a near-boundary population, not global probability calibration over all eligible events. The P2 mechanism grid is smaller still: 54 selected sites, six per cell, produce only two target transitions. Its null reorganization-KL association and absence of a resolved $O^B \times \mathcal{R}$ interaction are informative separation results, not evidence that no interaction exists in the underlying trajectory population.

The functional interventions remain at the last content token and evaluate the immediate next-token distribution; they are not interventions along a generated trajectory. Natural generation is available only for Thinking, is greedy, and is censored at 512 tokens for most examples. Only 35.9% emit an explicit thought- closing delimiter, so answer-stage estimates apply to a selected subset. Common- Trace Replay avoids token-alignment confounding by fixing the sequence, but this creates a different limitation: Base and A3B are evaluated under Thinking- generated content that they need not produce naturally. The replay therefore isolates checkpoint-associated routing response conditional on shared content; it is not a natural Base/A3B generation comparison, a causal training effect, or a

mechanism of reasoning accuracy. Gold problem-type labels, longer uncensored decoding, independent naturally generated controls, multi-token motifs, and trajectory-level interventions remain future work. Router scores are implementation coordinates: absolute levels and margins are reported only within an architecture. Qwen Common-Trace comparisons hold token identity fixed; Kanana, Nemotron, and Gemma4 hold prompt strings fixed but use checkpoint-specific tokenizers and normalized-position matching. Their evidence is therefore weaker than token-aligned replay and does not support exact site-by-site correspondence. The score-level decomposition uses shared-ID rank correlation and centered profile distance rather than rank-slot correlation, which would be tautologically one after sorting.

8 Conclusion

Post-training-associated MoE changes can occur below the resolution of aggregate expert utilization: under shared content, local score, rank, cutoff-neighborhood, and selected-set organization changes even when coarse usage remains similar. The tested global summaries do not recover this structure; in particular, functional correspondence does not imply routing correspondence, and the routing-optimal expert permutation is nearly identity with negligible held-out gain. Across four families, the decomposition is measurable but its direction and magnitude are heterogeneous. Functional relevance is distinct rather than automatic: normalized margin erosion prospectively targets routing transitions, and flip-conditioned restoration links expert replacement to output divergence, while reorganization itself is not a vulnerability score. Causal training attribution, task and layer transfer, and trajectory-level functional consequences remain open.

AI use statement

Generative AI tools assisted experimental code development, debugging, workflow documentation, analysis scripts, figure and table generation, and drafting and editing this manuscript. The authors reviewed AI-assisted outputs against saved configs and artifacts, executed numerical consistency checks, and retained negative results. No generative AI system is an author. The human authors take responsibility for the final content, claims, code, and artifacts. This disclosure must be updated before submission if subsequent uses change its scope.

Reproducibility statement

Supplementary code and artifacts specify checkpoint, pinned dataset revision, prompt and target-token protocol, split hashes, intervention definition, precision, seeds, and acceptance checks. One script regenerates every figure and result table from saved records. Appendices report derivative protocol lineage, negative gates, prospective selection hashes, routing-clamp validation, prefix-bootstrap trend intervals, and estimator uncertainty. The reserved confirmation set is excluded.

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A Supporting response geometry

For prefix x , let $p_0(y | x)$ and $p_\delta(y | x)$ be baseline and intervened next-token distributions. We define

$$s_i(x, y) = \left. \frac{\partial \log p_\delta(y | x)}{\partial \delta_i} \right|_{\delta=0}, \quad F = \mathbb{E}_{x, y \sim p_0(\cdot | x)} [s(x, y) s(x, y)^\top], \quad (8)$$

and accumulate $\hat{F} = N^{-1} \sum_n s_n s_n^\top$ in FP64. Within a fixed-routing region, $\mathbb{E}_x D_{\text{KL}}(p_0 \| p_\delta) = \frac{1}{2} \delta^\top F \delta + o(\|\delta\|_2^2)$. We compare the full predictor with its diagonal counterpart. The small pilot supports local quadratic prediction, but the full matrix does not beat the diagonal and pooled absolute magnitude fails the expanded-prefix robustness check. We therefore treat this geometry as supporting history rather than the paper’s main object.

B Natural identity turnover

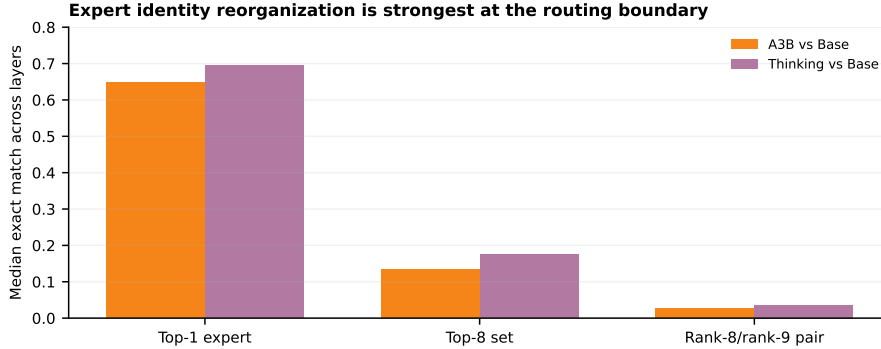


Figure 7: Median exact identity preservation across layers. The top-1 expert remains relatively stable, the full top-8 set usually changes, and the rank-8/rank-9 competing pair changes on more than 96% of natural question-token events.

C Generated-trajectory descriptive figure

D Implementation validation

Layer 47 was used for the formal finite-difference gate to minimize accumulated downstream roundoff; the same adapter and algebra are used at layer 23. An initial BF16 score path was rejected because its numerical discrepancy was too large; all reported response and intervention results use strict FP32.

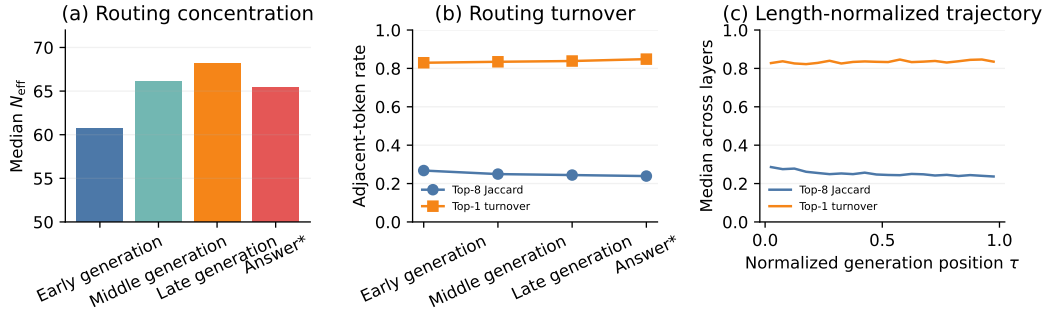


Figure 8: Routing along 64 Qwen Thinking generated trajectories. Effective expert count increases and adjacent-token top-8 overlap decreases from early to late observed generation. Most generations hit the 512-token cap; answer estimates use only delimiter-positive traces.

Table 7: Numerical and algebraic validation checks; all listed checks pass.

Check	Setting	Result
Zero intervention	layer 23, FP32	max logit difference 0
Inactive experts	layer 47, 20 cases	autograd and FD max 0
Finite difference	layer 47, 80 active	median error 0.735%
Finite difference	layer 47, 80 active	P90 error 4.942%
Finite difference	layer 47, 80 active	sign agreement 100%
Repeat score	fixed seed/prefix	max difference 0
F symmetry	$N = 512$	relative error 0
Streaming vs. offline	$N = 512$	1.57×10^{-16}
Positive semidefinite	$N = 512$	$\lambda_{\min} = -2.24 \times 10^{-20}$

E Detailed local prediction results

Negative R^2 for small subsets reflects tiny observed KL and only 10 cases per stratum. Full and diagonal are identical for $|S| = 1$ by construction.

F Routing analysis details

G Sampling stability and estimator redesign

H Artifact scope

All figures are generated from saved JSONL, JSON, CSV, and NumPy artifacts. The pilot local prediction unit is an intervention mean over 16 prefixes; the E4 expanded unit is an intervention mean over 128 prefixes; the routing unit is one intervention–prefix record. These units remain separate to avoid inflating sample size. E4 artifacts are stored under artifacts/followup/e4_expanded_local_v2/ and include the 20,000- replicate bootstrap results. The generation manifest records output hashes and that the redesign confirmation set was not accessed. The Qwen–A3B and Kanana score-replay trend intervals are generated from artifacts/graph/reorganization_trend_bootstrap_n64.json using 20,000 prefix resamples; no additional model forward is required.

I Derivative protocol lineage

The original stock sequence-wide G0 full-vector finite-difference gate failed. Component audits localized the discrepancy to the stock in-place expert aggregation: torch.func.jvp and low-level forward AD agreed with each other but not with reverse mode or stable finite

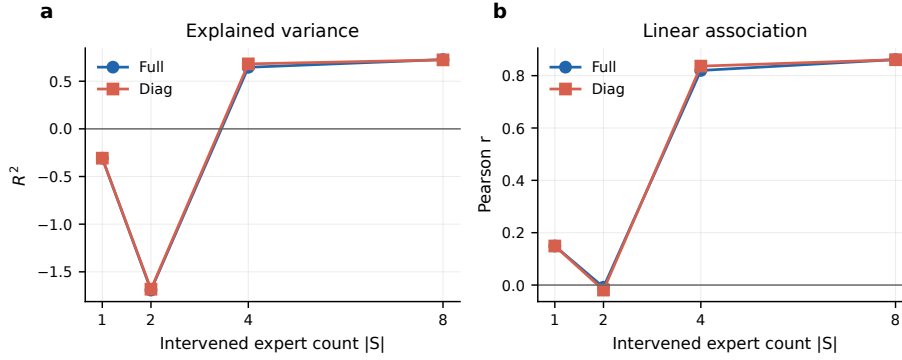


Figure 9: Descriptive local metrics by intervention subset size. Each stratum contains 10 interventions; the pooled result is primary.

Table 8: Local prediction stratified by intervened expert count.

$ S $	Predictor	n	R^2	Pearson r	Spearman ρ_s
1	Full	10	-0.308	0.149	0.423
1	Diagonal	10	-0.308	0.149	0.423
2	Full	10	-1.691	-0.008	0.067
2	Diagonal	10	-1.682	-0.019	0.067
4	Full	10	0.646	0.820	0.503
4	Diagonal	10	0.682	0.836	0.503
8	Full	10	0.727	0.861	0.903
8	Diagonal	10	0.724	0.861	0.879

differences. On the documented out-of-place index_add amendment, forward AD, low-level dual tensors, autograd.functional.jvp, reverse mode, and finite differences agree at the audited router event. Full-vector forward/reverse relative error is 1.67×10^{-6} with cosine effectively one; stable $\epsilon = 10^{-3}$ finite difference has relative error 0.0129 and cosine 0.99992.

A separate 8-prefix primal audit verifies that the amendment changes only expert aggregation, not parameters, routing policy, intervention definition, targets, or thresholds. Small FP32 differences are reported as accumulation-order effects rather than requiring bit identity. The stock failure remains recorded; it is not retroactively relabeled. The amended scalar router-margin derivative subsequently passes its primary AD oracle gate. Position-dependent full-vector FD remains an explicit numerical diagnostic, not a replacement gate.

J Event-level routing and clamp details

E5-A logs 2,206,080 downstream events for 120 interventions and 128 prefixes, forming 15,360 record aggregates. It finds 551 flip events in 225 records and 68 interventions. No flip satisfies $2\|\Delta q\|_\infty < \gamma$. Among affected records, 145/225 flips occur at the record’s minimum-margin event. Flip-event median margin is 0.003109 versus 0.091072 for nonflips; median router displacement is 0.03374. Flips concentrate in layers 45–46, followed by 42, 41, and 44. Event maxima reproduce legacy record-level ρ exactly.

First- and all-clamp KL are exactly equal in 216 records. The nine exceptions are retained. The minimum all-clamp suppression is -0.0106 ; one slightly negative case prevents an absolute restoration claim.

K Prospective selection and cross-architecture details

Generated trajectory and Common-Trace Replay audit. The Thinking trajectory set contains 64 unique prompts, 32,344 generated tokens, and 36,853 prompt-plus-continuation to-

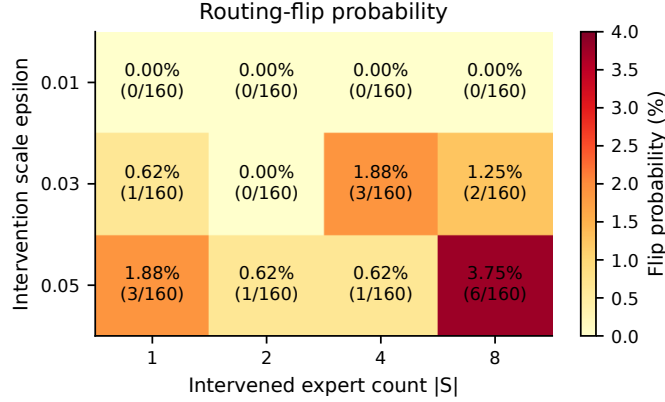


Figure 10: Flip probability by perturbation scale and subset size. Each cell contains 160 records. No record flips at $\epsilon = 0.01$. Percentages are descriptive because records share prefixes and interventions.

Table 9: Record-level routing-transition summary.

Quantity	Estimate	Uncertainty/reference
Records	1,920	16 prefixes $\times$ 120 interventions
Records with flip	17	cluster rate CI [0.0021, 0.0203]
Affected interventions	14	of 120
ρ ROC-AUC	0.9977	Mann-Whitney $p = 4.24 \times 10^{-16}$
Median ρ , flip	33.415	no-flip P90 = 0.892
Mean KL, flip	6.822×10^{-4}	no-flip = 2.765×10^{-9}
Mean log-KL contrast	flip minus no-flip	cluster CI [6.77, 9.62]

kens. Only 6/64 traces finish before the 512-token cap and 23/64 contain the closing thought delimiter. The replay grids for Base, A3B, and Thinking each contain the same 64 complete sequences and pass record-count, uniqueness, shape, normalized-weight-sum, and content-hash checks. Their record SHA256 values begin 43f83e250209bced, ddfcd7ea2520f97e, and d2a7c4b9eccc9558. No sealed confirmation record is read. Exact rank-8/rank-9 agreement with Base is .022/.022/.022 across A3B’s early/middle/late segments and .016/.015/.014 for Thinking. We retain these values for completeness but prefer rank-6:10 Jaccard because exact pair identity is a high-cardinality statistic.

P2 reorganization-risk pilot audit. The baseline phase evaluates 1,536 candidate sites (64 traces, one stratified generation position, and 24 downstream layers), then freezes 54 interventions with six in each $O^B \times \mathcal{R}$ cell. Reorganization cut points are .571 and .750; log-risk cut points are -2.064 and -1.506 . Selection and outcome SHA256 values begin 0abeb788ec5e116e and 679609fbb464bad8. All selection-hash, active-source, baseline-repeat, target-route, finite-value, and confirmation-isolation checks pass. The 54 sites span 40 prefixes; prefix reuse across different cells is retained in the cluster bootstrap.

P3 score replay audit. The score replay stores 36,853 prompt-plus-continuation tokens for each of Base, A3B, and Thinking, with 48 layers, top-16 IDs, and top-16 FP32 router scores. Each score stream passes record-count, uniqueness, shape, descending-score, normalized-weight, content-hash, and confirmation-isolation checks. The score record SHA256 values begin 7fff8f83b37db70b, 860f492e19ec66b0, and 589a29b9d0004042; the analysis summary is stored under artifacts/graph/common_trace_score_decomposition_n64/. The permutation-control summary is stored under artifacts/graph/expert_permutation_control_qwen_n64/. Its Hungarian map is fit separately for every layer on 32 alternating prefixes and evaluated on the other 32, then the folds are reversed. Fingerprint channels are early/middle/late top-16

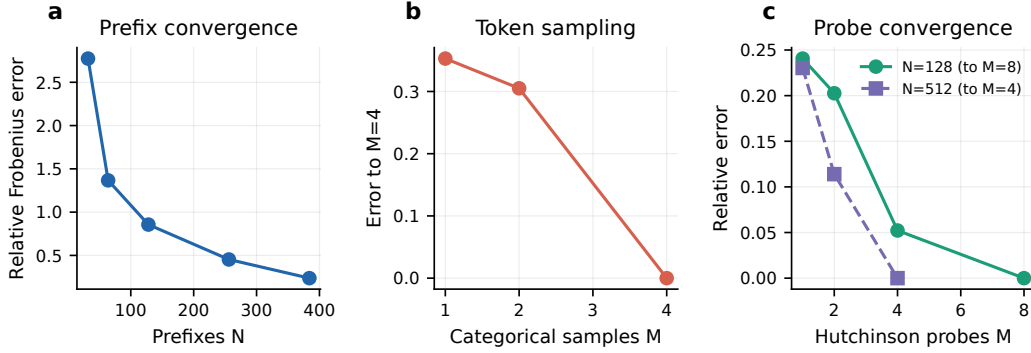


Figure 11: Sampling diagnostics: nested-prefix estimates, categorical token sampling, and Hutchinson probe convergence. Probe convergence does not imply convergence over independent prefixes.

Table 10: Prefix-split stability; lower difference and higher overlap are better.

Estimator	N	Half difference	Top-8 overlap	Decision
Categorical $M = 4$	512	0.762	0.613	unstable
Categorical RB $M = 4$	128	0.953	0.693	unstable
Hutchinson $M = 8$	128	0.892	0.564	unstable
Hutchinson $M = 4$	512	0.566	0.631	narrow gate fail

score, top-16 presence, top-8 activation, and selected gate weight. All reported Jaccard changes use held-out prefixes and 20,000 prefix-bootstrap draws. The route-optimized control is stored under artifacts/graph/route_optimized_permutation_qwen_n64/. It uses the same folds and learns each layer’s assignment from binary top-8 activation vectors over 16,126 or 16,218 training-context token sites; no modified-forward outcome enters either assignment. For Thinking–Base, the two folds improve the mean training activation-Jaccard objective by .00047 and .00052 over identity; their non-fixed fractions are .123 and .121.

Full SHA256 values are stored in the released manifests. For compact audit, their leading 16 hexadecimal characters are: first G4 prefix/selection/outcome e45109de55c77ec, 287d825c913ed2ff, and be8aa423eee30525; second G4 prefix/selection/outcome 114a664f7a1131e4, 490d7ba687b3e948, and 650783637ac75daf. A3B selection/outcome begin bb615e3a3f6b8ab5/2478bcbdde1836e2; Thinking begins 1dc57f6e46fdb8f5/6377ee7825d3be1f. Corrected-score Nemotron Base selection/outcome begin 7b72886c99438242/ c3bfe808238a4ba9; Nemotron Post begins e1fc1b80a63d2626/fb89b604a1164e44. All baseline repeats, counts, target boundaries, finite values, and selection hash checks pass. No confirmation data are used.

P0 prospective freeze. The new P0 prefix, component-selection, component-outcome, non-max-selection, and non-max-outcome SHA256 values begin 58969e9184dc3819, f80c92e39385f701, 54b9ad4bfff0117a5, 7d87e7a7763aff6d, and e0a44cef3471313f. The two 64-prefix subsets are disjoint from both earlier prospective G4 splits. Both outcome programs verify the frozen selection hash before loading the model and never recompute selection features.

For 128 paired Base–Thinking prefixes, selected-event γ is larger for Thinking in 73 cases (57.0%) and smaller in 55; the paired median ratio is 1.44. The median derivative over all eight active sources is smaller in 72 cases (56.2%) and larger in 56; its paired median ratio is 0.83. Hence the aggregate margin shift is statistically resolved but heterogeneous, not pointwise dominance. The stored basis contains all eight source directions only at the

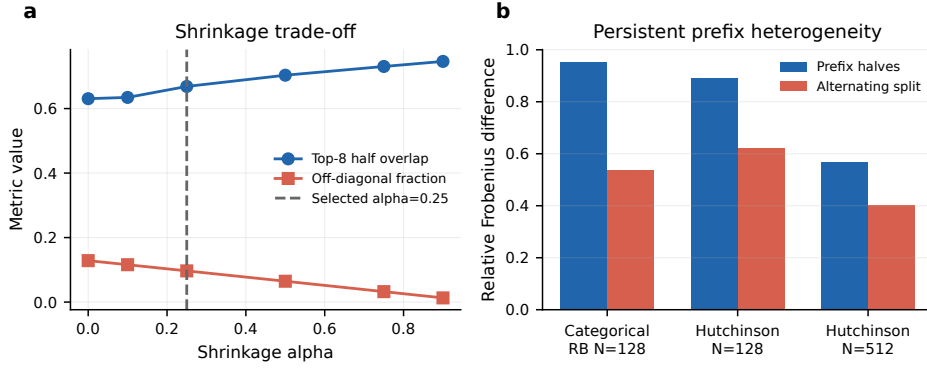


Figure 12: Exploratory redesign. Shrinkage trades subspace stability against off-diagonal magnitude, while prefix heterogeneity persists across estimators. Outcome tuning did not select a superior estimator, so confirmation stayed sealed.

Table 11: Exploratory no-flip shrinkage tuning smoke ($n = 11$).

Predictor	R^2	MSE	MAE
Shrinkage ($\alpha = 0.25$)	0.5042	4.1769×10^{-19}	4.5973×10^{-10}
Raw response	0.5057	4.1644×10^{-19}	4.5792×10^{-10}
Diagonal	0.4991	4.2201×10^{-19}	4.6515×10^{-10}

selected target event; it cannot recover the unconditional distribution over every nonselected eligible event.

Common-site boundary turnover. Freezing the Base-selected physical (ℓ, t) does not freeze the underlying decision boundary. Qwen Thinking preserves the Base rank-boundary expert pair in only 1/128 prefixes, and Kanana Thinking in only 2/128. At those physical sites, the state-specific Thinking/Base geometric-mean ratios are 40.5 [27.4,60.2] and 52.0 [35.9,75.4] for γ , 0.34 [0.28,0.41] and 0.29 [0.23,0.38] for maximum active-source $|g|$, and 0.0083 [0.0053,0.0129] and 0.0056 [0.0036,0.0087] for maximum $|g|/\gamma$. Because the boundary expert pair changes in nearly every prefix, these are state-specific physical-site operating-point contrasts, not same-boundary geometric effects.

Nemotron amendment and superseded path. Nemotron routes by $r_i = \sigma(q_i) + b_i$, not by raw logits. We therefore define its rank-6/rank-7 margin in corrected score space. An initial raw-logit-boundary analysis is retained only as a superseded diagnostic and is not reported as a scientific result. The corrected protocol independently freezes all 128 target events before modified forwards, perturbs only routed experts (not the shared expert), and uses a central finite-difference derivative with step 10^{-3} . Both corrected 384-record outcome grids pass selection-hash, uniqueness, finite-value, exact final-logit repeat, and exact target-boundary checks. Maximum absolute selection-to-outcome margin drift is 3.81×10^{-6} for Base and 8.39×10^{-5} for Post, while boundary expert identities remain exact.

Nemotron high-risk flip rates over ascending within-model $\mathcal{R}$ quintiles are [0,0,0.038,0.32,0.84] for Base and [0,0,0,0.28,0.84] for Post. These bins are diagnostic within-model calibration, not a shared probability scale. Their sharp ordering complements the cross-family functional summary: the risk primitive transfers even though the Post/Base effect and geometry contrasts are unresolved. A multi-step FD audit at eight frozen events fails for both Nemotron states: relative to 10^{-3} , median relative L2 errors are 1.00 at 3×10^{-4} and 0.66–0.71 at 3×10^{-3} , with low sign agreement. The discrete derivatives are FP32 cancellation-limited. This negative numerical result does not alter the already frozen 10^{-3} prospective enrichment, but it rules out strong cross-step claims about Nemotron’s derivative magnitude or checkpoint shift.

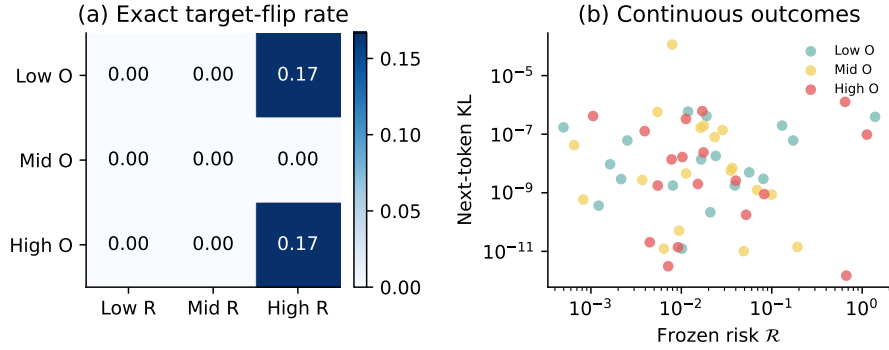


Figure 13: Balanced selected-site mechanism pilot. Exact target-flip rates and next-token KL after crossing rank-6:10 reorganization with frozen Base risk. Only two target flips occur, both at high risk but at opposite reorganization levels; the experiment does not establish an $O^B \times \mathcal{R}$ interaction.

Table 12: Routing restoration over all 225 observed transition records.

Clamp	Median suppression	P10	Fraction > 0.5	Fraction > 0.9
First changed event	0.99677	0.85672	0.9600	0.8756
All changed IDs	0.99677	0.86761	0.9689	0.8844
All IDs and weights	0.99661	—	—	—

Kanana architecture and derivative-associated variation. Kanana-2 uses a DeepSeek-V3-style MLA/MoE implementation with 48 decoder layers: layer 0 is dense and layers 1–47 are MoE. Each MoE layer has 128 routed experts, top-6 selection, and one shared MLP whose width equals two routed experts. We use physical source layer 23 and destinations 24–47, intervene only on routed contributions, and leave the shared MLP untouched. Expert identities are selected by $\sigma(q) + b$; contribution weights instead gather uncorrected sigmoid scores, renormalize over top-6, and multiply by 2.448. This explicitly separates ranking score from contribution weight. Stock/out-of-place aggregation relative L2 is 6.18×10^{-8} for Base and 6.04×10^{-8} for Thinking.

Base and Thinking selection/outcome hashes begin e3d6e497867da070/df86a9ca7703c59e and 3951599cfa5elf25/6da1b7ab6dc96a71. Each checkpoint uses a distinct 512-prefix atlas and a pre-outcome frozen selection. Both 384-record grids pass exact inactive/zero-delta, baseline repeat, boundary, finite-value, uniqueness, and selection-hash checks, with no relaxed matches. Ascending-risk flip rates are $[0, .038, 1, 1, .92]$ and $[0, 0, .231, .92, .96]$. The nonmonotonic top-bin fluctuation is compatible with 25–26 prefixes per bin; the defensible claim is sharp low-to-high stratification, not deterministic calibration.

Both checkpoints use the identical plain prompt rather than the Thinking chat template. Consequently the comparison is cleaner with respect to input context, but remains an observational weight-state association and does not evaluate generated reasoning. A 12-event follow-up varies the central-FD step over 3×10^{-4} , 10^{-3} , and 3×10^{-3} . Relative to 10^{-3} , median relative L2 errors are 0.055/0.010 for Base and 0.021/0.008 for Thinking; sign agreement and active-source Spearman are 1.0 throughout. The derivative stability gate therefore passes, closing the step-size concern while leaving the selected-event comparison limitation intact.

Gemma architecture and mode-conditioned context. The executable Gemma 4 26B-A4B model has 30 MoE text layers, 128 routed experts, top-8 selection, and no shared expert. Its router first computes FP32 softmax probabilities, selects top-8 identities, then renormalizes selected weights and applies per-expert scales. We therefore define γ as the rank-8/rank-9 gap in the pre-selection softmax probabilities. Stock in-place and derivative-safe out-of-place aggregation differ by relative L2 7.09×10^{-8} . All three 512-prefix atlases and 384-record

Table 13: Held-out functional-fingerprint permutation control. Entries show raw Jaccard, Jaccard after cross-fitted Hungarian relabeling, and matched-minus-raw change with 95% prefix-bootstrap CI (20,000 draws).

Pair	Set	Raw	Matched	Δ (95% CI)
A3B/Base	Top-8	.655	.437	-.218 [-.222, -.215]
	Rank-6:10	.348	.213	-.135 [-.138, -.132]
Thinking/Base	Top-8	.613	.366	-.247 [-.251, -.243]
	Rank-6:10	.298	.163	-.135 [-.139, -.132]
Thinking/A3B	Top-8	.672	.442	-.230 [-.234, -.225]
	Rank-6:10	.366	.218	-.148 [-.152, -.145]

Table 14: Prospective non-max ordering conditional on near-boundary candidates. One downstream event is sampled per prefix before outcomes; all eight active source directions are retained. Intervals are prefix-cluster bootstrap 95% CIs.

Risk quintile	n	Frozen $\mathcal{R}$ range	Exact target flip
Q1	103	$[2.0 \times 10^{-6}, 7.8 \times 10^{-4}]$	.000 [.000,.000]
Q2	103	$[8.2 \times 10^{-4}, 2.9 \times 10^{-3}]$	.000 [.000,.000]
Q3	102	$[2.9 \times 10^{-3}, 8.1 \times 10^{-3}]$	.000 [.000,.000]
Q4	102	$[8.1 \times 10^{-3}, 4.0 \times 10^{-2}]$	.000 [.000,.000]
Q5	102	$[4.0 \times 10^{-2}, 3.20]$	.118 [.024,.219]

outcome grids pass inactive, zero-delta, repeat, boundary, finite-value, and selection-hash checks; no matched-low selection is relaxed.

Eight-event multi-step FD audits also pass for Base, IT-OFF, and IT-ON. Relative to the frozen 10^{-3} step, median relative L2 errors at 3×10^{-4} are 0.115, 0.102, and 0.070, and at 3×10^{-3} are 0.028, 0.034, and 0.024; sign agreement is 1.0 throughout and median active-source Spearman is at least 0.964. Thus all reported Gemma conditions pass the same numerical stability gate.

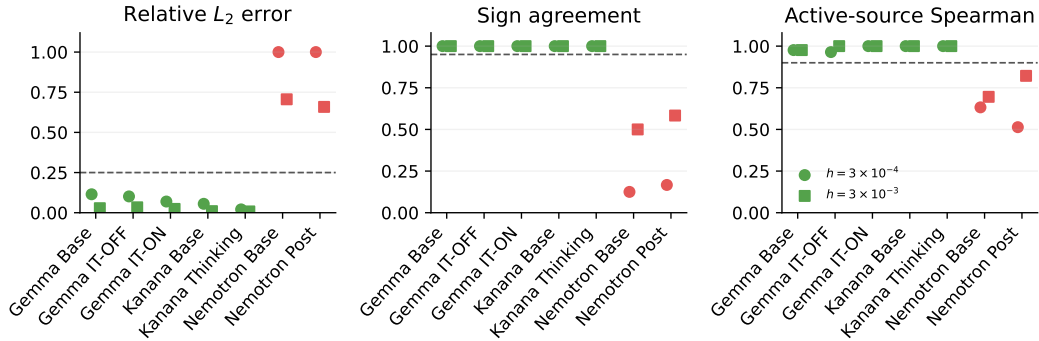
Gemma Base, IT-OFF, and IT-ON high-risk flip rates over ascending $\mathcal{R}$ quintiles are $[0, 0, 0, 0, 0.76]$, $[0, 0, 0.115, 0.92, 0.96]$, and $[0, 0, 0, 0.36, 0.96]$. OFF and ON use identical IT weights but distinct official chat renderings. In an audited prefix they contain 68 and 71 tokens and share no terminal suffix: OFF appends an empty thought block, whereas ON prepends a system turn with $\langle |think| \rangle$. Accordingly, the resolved ON/OFF geometry shift is evidence for mode-conditioned context dependence, not an isolated causal effect of thinking.

Table 15: Cross-checkpoint selected-event geometry and local susceptibility.

Checkpoint	Median γ	Median $ g $	Median $ g /\gamma$	Random flip	High/low
Base	0.00229	0.0787	30.44	0.1172	4.54×
A3B	0.00174	0.0664	29.00	0.1250	4.31×
Thinking-2507	0.00305	0.0547	15.91	0.0703	5.00×

Table 16: Qwen source-layer utilization and functional identity across the checkpoint progression. All utilization statistics use the same 512-prefix build set; source-match rates use the 128-prefix prospective split.

State	Entropy	N_{eff}	Top-16 Jaccard vs. Base	High-risk source match
Base	3.455	31.7	1.000	1.000
A3B	3.490	32.8	.882	.266
Thinking	3.445	31.3	.684	.281

Figure 14: Multi-step finite-difference audit relative to the frozen $h = 10^{-3}$ oracle. Gemma and Kanana conditions pass the preregistered relative error, sign, and rank gates at 3×10^{-4} and 3×10^{-3} ; both Nemotron states fail. Green/red indicate the overall condition-level decision; dashed lines are acceptance thresholds.Table 17: Multi-step derivative audit. Each pair reports the median at $h = 3 \times 10^{-4}$ / 3×10^{-3} relative to the frozen 10^{-3} estimate. Spearman and sign agreement are over active source directions.

Condition	Relative L_2	Sign agreement	Spearman	Decision
Gemma Base	.115/.028	1.00/1.00	.976/.976	Pass
Gemma IT-OFF	.102/.034	1.00/1.00	.964/1.00	Pass
Gemma IT-ON	.070/.024	1.00/1.00	1.00/1.00	Pass
Kanana Base	.055/.010	1.00/1.00	1.00/1.00	Pass
Kanana Thinking	.021/.008	1.00/1.00	1.00/1.00	Pass
Nemotron Base	1.00/.706	.125/.500	.632/.696	Fail
Nemotron Post	1.00/.659	.167/.583	.514/.822	Fail